

Empire’s Taste: A Medallion Data Pipeline for Gastronomic Trend Analysis and Sentiment Intelligence in NYC

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Abstract—Food selection is a critical component of event planning in highly competitive gastronomic environments such as New York City, where chefs and catering managers must align menu decisions with current culinary trends and guest preferences, a task still largely driven by intuition rather than systematic analysis of public food discourse. This work presents Empire’s Taste, an automated data pipeline that daily harvests culinary news from Eater NY, restaurant reviews from OpenTable via web scraping, and queries the Spoonacular recipe API using trends identified through natural language processing, organizes data across three progressive quality layers orchestrated with Apache Airflow, and delivers results through interactive dashboards and web applications. The deployed system answers three key questions for event planning professionals: what dishes and ingredients are trending in NYC right now, how the public feels about specific foods and cuisines, and which recipes best match those trends while accommodating dietary restrictions, providing chefs, catering managers, and event planners with a reliable, evidence-based foundation for menu decisions.

Index Terms—Sentiment analysis, data pipeline, medallion architecture, food trends, Apache Airflow, NLP, recipe recommendation

I. INTRODUCTION

Menu design for events is one of the most consequential decision-making challenges in the hospitality industry. In a city like New York, where thousands of restaurants, luxury hotels, and catering services organize corporate dinners, weddings, and cultural celebrations, selecting the right dishes for a specific audience is both commercially critical and difficult. Despite this importance, menu planning has historically depended on the intuition of experienced chefs and catering managers rather than on systematic analysis of public food discourse. There is an enormous and continuously updated body of culinary media — news articles, editorial reviews, food blog coverage — that reflects what dishes, ingredients, and cuisines are currently capturing public attention, but this information is unstructured and difficult to translate into actionable guidance for event professionals. At the same time, structured recipe databases catalog thousands of dishes with detailed nutritional, dietary, and preparation metadata. The challenge lies in bridging these two worlds to answer the questions that matter most to a catering manager: what is

trending right now, and which recipes best fit those trends for a specific event and audience.

Prior work has addressed parts of this problem from several directions. In the area of opinion mining, Hu and Liu [1] introduced lexicon-based methods for extracting feature-level opinions from customer reviews, establishing the foundation for understanding how specific food attributes are perceived by the public. Pang and Lee [2] later formalized the field of sentiment analysis with a comprehensive survey covering lexicon-based scorers, Naive Bayes classifiers, and Support Vector Machines, enabling systematic opinion classification across domains including food and restaurants. More recently, Yadav and Vishwakarma [3] demonstrated that deep learning architectures — particularly convolutional and recurrent neural networks — achieve significant accuracy improvements over rule-based sentiment methods, though at considerably higher computational cost.

In the food recommendation domain specifically, Minct et al. [4] showed that performing sentiment extraction at the dish level, rather than aggregating scores at the restaurant level, substantially improves recommendation precision, motivating fine-grained NLP analysis of culinary content. Zhang et al. [5] proposed neural models that jointly learn from unstructured review text and structured recipe metadata, demonstrating that combining heterogeneous sources captures complementary signals that neither alone can provide. Burke [6] established the framework for hybrid recommender systems merging collaborative filtering, content-based reasoning, and knowledge-based constraints, while Adomavicius and Tuzhilin [7] showed that context-aware systems — which factor in situational variables such as event type or group composition — produce meaningfully more relevant suggestions, a result that directly motivates event-specific menu recommendation.

Despite this body of work, existing systems address sentiment analysis or recipe recommendation in relative isolation and rarely integrate both within a production-grade pipeline that tracks how public food preferences evolve and uses that signal to drive recipe discovery. This paper presents Empire’s Taste, a fully implemented system that closes this gap. The pipeline daily ingests culinary news from Eater

NY, restaurant reviews from OpenTable via web scraping, and applies natural language processing — using Pandas [13] for data transformation and spaCy [12] for semantic filtering — to identify trending food topics and dynamically query the Spoonacular recipe API [9]. Data flows through a three-layer Medallion architecture [11] orchestrated with Apache Airflow [10], with public sentiment scored using the VADER model [8] and large-scale aggregations computed via PySpark. Results are delivered through interactive dashboards and a web application, giving chefs and event planners direct answers to three practical questions: what is trending in NYC’s food scene right now, how does the public feel about specific dishes and cuisines, and which recipes best match those trends while accommodating the dietary requirements of their guests.

II. METHODS AND MATERIALS

A. System Overview and Design Rationale

Empire’s Taste is designed as a modular, end-to-end data pipeline that transforms unstructured culinary media and diner reviews into actionable gastronomic intelligence. The system uses a three-layer Medallion-style organization — Bronze, Silver, and Gold — where each layer has a well-defined responsibility: raw ingestion, cleansing and normalization, and analytical aggregation respectively. This layered separation isolates failures, enables independent reprocessing of any layer without discarding earlier work, and supports the accumulation of historical partitions without overwriting prior runs. Figure 1 illustrates the overall data flow from Eater NY, OpenTable, and Spoonacular through the three layers to the serving layer. Apache Airflow was selected as the orchestration engine because it provides declarative directed acyclic graphs (DAGs), built-in retry and sensor primitives, and a mature operational UI, all critical for a pipeline designed to run as a controlled manual workflow during the final deployment stage.

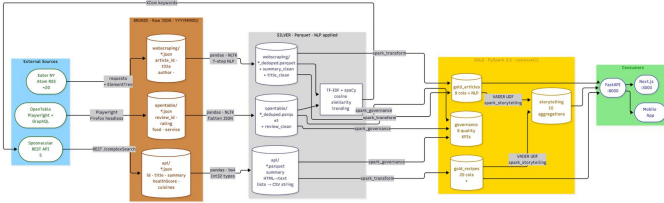


Fig. 1. High-level architecture of the Empire’s Taste pipeline, showing the Bronze, Silver, and Gold data layers orchestrated by Apache Airflow, and the downstream serving components.

B. Data Ingestion — Bronze Layer

The pipeline begins with three complementary ingestion sources that together capture editorial discourse, diner opinion, and structured recipe knowledge. The first source is the Eater NY RSS/Atom feed, harvested by a web scraper that retrieves up to twenty articles and stores each as a timestamped JSON document under a date-partitioned directory. OpenTable was added as a final data-source extension after early sentiment results from Eater NY proved excessively positive, because editorial food coverage produced little negative or neutral signal

TABLE I
DATA LAYER SUMMARY

Layer	Format	Partition Layout	Stored Content
Bronze	JSON	datalake_bronze/ api/ YYYY-MM-DD/ webscraping/ opentable/ YYYY-MM-DD/	Raw recipe API responses, article payloads, and OpenTable review JSON files.
Silver	Parquet	datalake_silver/ api/ YYYY-MM-DD/ webscraping/ opentable/ YYYY-MM-DD/	Cleaned articles, recipes, and reviews, stored as Parquet files per source.
Gold	Parquet	datalake_gold/ YYYY-MM-DD/	Consolidated outputs for articles, recipes, reviews, governance, and storytelling.

for downstream dashboards. This second source is collected with a headless Playwright and Firefox scraper that obtains a browser session automatically, calls the internal GraphQL review endpoint, and paginates up to five pages of ten reviews for each restaurant. Each OpenTable record preserves the review identifier, free-text comment, overall, food, service, and ambience ratings, dining date, restaurant slug, and user metro. The third source is the Spoonacular recipe API, queried with dynamically extracted keywords rather than a fixed ingredient list, a design decision that makes recipe discovery trend-driven rather than static. Raw responses are written using the same YYYY-MM-DD partitioning convention, and JSON is preserved at this layer without transformation so downstream processing decisions can be revisited. Table I summarizes the data lake layout, formats, and responsibilities of each layer following the project directory structure.

C. Trend-Driven Keyword Extraction — Silver Layer

The Silver layer performs three distinct operations on the ingested data. First, it applies NLP preprocessing to scraped articles and OpenTable reviews: text is lowercased, punctuation is stripped, and English stopwords are removed using NLTK, producing clean text fields stored in Parquet format for columnar efficiency. Second, a TF-IDF vectorizer (scikit-learn) is applied to the combined article-review corpus to extract the top-ten trending food keywords from both editorial and diner-generated text. Additional review stopwords, such as generic praise and service terms, prevent non-gastronomic words from driving recipe search. These keywords are passed downstream via Airflow XCom to the Spoonacular ingestion task, which then queries the API specifically for recipes matching those terms. This feedback mechanism is the central architectural decision that distinguishes Empire’s Taste from a static pipeline: the recipe corpus is shaped by both culinary press coverage and real diner reviews. For the recipe branch of the Silver layer, list columns are flattened to comma-separated strings, HTML is stripped from summaries using BeautifulSoup, and records are deduplicated by identifier before writing to Parquet. Figure 2 shows the Airflow DAG task sequence, including the OpenTable extraction and cleaning steps and the XCom wire connecting trend extraction to recipe ingestion.

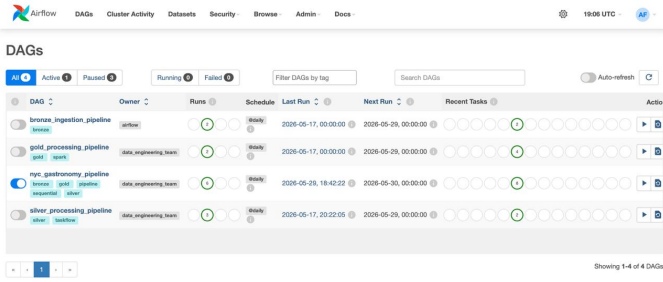


Fig. 2. The `nyc_gastronomy_pipeline` DAG in the Airflow UI, showing the unified task sequence from Eater NY and OpenTable ingestion through trend extraction, Spoonacular search, Gold preparation, S3 upload, and cleanup.

TABLE II
GOLD LAYER DAILY OUTPUTS

File Prefix	Content	Consumer
gold_recipes	Deduplicated recipe records joined with trend context	API, web, mobile
gold_articles	Cleaned articles enriched with sentiment scores	API, dashboards
gold_reviews	Cleaned OpenTable reviews with ratings and sentiment	API, dashboards
storytelling	Trend, keyword, and sentiment aggregations	Storytelling dashboard, API
governance	Data quality KPIs by source and field	Governance dashboard, API

D. Aggregation and Sentiment Scoring — Gold Layer

The Gold layer is implemented with PySpark to handle the cross-source joins and the multiple aggregation passes required to produce analytics-ready outputs. Each Spark task opens and closes its own `SparkSession` independently to avoid conflicts with concurrent JVM instances, a constraint enforced at the DAG level via `max_active_runs=1`. The transformation job now produces dedicated article, recipe, and review outputs, while the storytelling job applies the VADER sentiment model to both `article_summary_clean` and `text_clean`. Its aggregations include article sentiment distribution, review sentiment distribution, weekly compound-score trends, keyword frequency with associated sentiment polarity, and entity-level summaries stored in a flat schema for direct API consumption. The governance script computes data quality KPIs including null rates per field, duplicate rates, IQR-based outlier rates, and schema compliance. Table II lists the Gold output files produced inside each daily partition and their downstream consumers. All outputs are written as Spark coalesced single-file Parquet directories, and a post-processing task uploads the latest partition to Amazon S3 for production deployment.

E. Serving Layer

A FastAPI application reads from the Gold layer through a loader abstraction that supports both local filesystem access for development and S3 access for production, resolving the most recent date partition automatically so the API always serves the latest available data. Three router modules ex-

pose governance KPIs, storytelling aggregations, and trend-aligned recipe recommendations through a documented REST interface. The storytelling sentiment endpoint now combines article sentiment with OpenTable review sentiment, and a new sources endpoint returns article counts, article links, review counts, and review volume by restaurant. Two Plotly Dash dashboards consume these endpoints and auto-refresh every five minutes: the Storytelling dashboard is oriented toward culinary professionals and presents sentiment trends, keyword analysis, and source coverage as shown in Figure 3; the Governance dashboard is oriented toward the data engineering team and presents pipeline health, missing-data severity, and schema quality as shown in Figure 4. A Next.js web application and a React Native mobile client built with Expo SDK 54 provide interactive front-ends for end users, both relying on the same FastAPI contract.

Empire's Taste — NYC Food Trends

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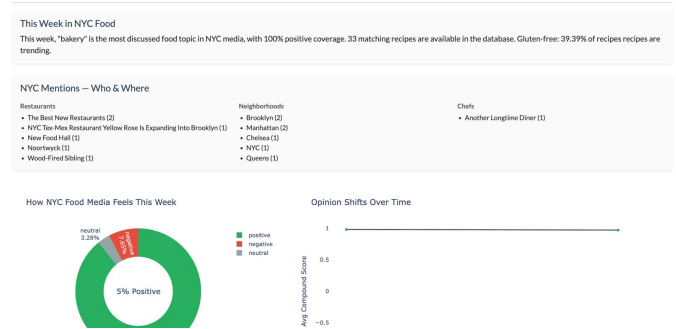


Fig. 3. The Storytelling Dash dashboard displaying the NYC Food Mood Index, sentiment trend over time, top food keywords, and source-aware sentiment derived from the Gold layer aggregations.

Empire's Taste — Data Governance Dashboard

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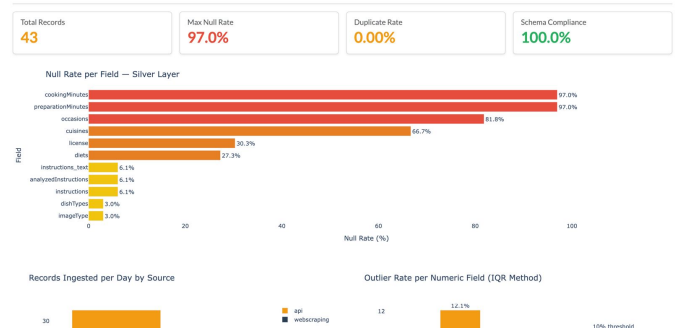


Fig. 4. The Governance Dash dashboard displaying pipeline health, data quality indicators, missing-data severity, and source-level monitoring for the Bronze, Silver, and Gold layers.

F. Assumptions and Limitations

The system still depends on the availability and structure of third-party web sources, especially the OpenTable GraphQL endpoint and the browser session required by the headless scraper. The VADER model may underperform on longer

editorial prose or nuanced restaurant reviews, and no domain-specific fine-tuning was applied to culinary vocabulary. The five-second HTTP timeout used across API clients is a pragmatic choice that may cause spurious failures on slow networks; a retry strategy with exponential backoff would improve robustness in production. Finally, because the DAGs are configured for manual execution, the Gold layer reflects the latest triggered run rather than a guaranteed daily rolling window.

III. RESULTS & DISCUSSION

A. Pipeline Execution

The unified DAG now executes the complete sequence `extract_eater_ny`, `clean_eater_ny`, `extract_opentable`, `clean_opentable`, `analyze_trends`, `search_spoonacular`, `clean_recipes`, `prepare_gold`, `upload_s3`, and `cleanup`. The XCom mechanism successfully passes trending keywords from the combined article-review analysis to the Spoonacular search task, enabling trend-driven recipe queries. The DAGs were configured with manual triggering to avoid unintended reruns during workshop validation. Table III summarizes representative outputs from a production run.

TABLE III
REPRESENTATIVE PIPELINE OUTPUTS FROM A DAILY RUN.

Stage	Output	Details
Bronze Web	20 articles/run	Deduplicated by <code>article_id</code> before saving
Bronze Reviews	Up to 50 reviews/restaurant	OpenTable GraphQL pages collected through a headless browser session
Bronze API	5 recipes/keyword	Queried per trending term extracted by NLP
Silver Text	Articles and reviews	Clean text fields generated for TF-IDF and sentiment scoring
Silver API	1 Parquet/keyword	List columns flattened; HTML stripped
Gold	5 Parquet outputs	Recipes, articles, reviews, governance, and storytelling

B. Trend Keyword Extraction

The hybrid TF-IDF and semantic filtering approach successfully discriminates gastronomic terms from contextual noise across both article summaries and diner reviews. For a batch covering restaurant openings and user feedback, naive frequency counts surfaced operational or generic opinion words such as *service*, *staff*, *amazing*, and *dinner*. After adding review-specific stopwords and semantic filtering, the pipeline returned terms such as *indian*, *coffee*, *sandwiches*, *italian*, *cocktail*, *pizza*, and *sushi* — terms directly relevant for recipe querying. The final keyword board normalizes TF-IDF scores to a 0–100 range and translates them into human labels such as *Highly trending*, *Trending*, *Growing*, *Mentioned*, *Mixed reviews*, or *Neutral coverage*, improving interpretability for non-technical users.

C. Sentiment Analysis

VADER sentiment classification was extended from article summaries to OpenTable review text. This change addressed

the earlier positive-bias problem, where the storytelling endpoint only consumed Eater NY article sentiment and produced a nearly all-positive distribution because the source mostly reflects promotional editorial coverage. OpenTable reviews were therefore introduced at the end of development to add direct diner feedback and expose negative or neutral experiences that were absent from the original source mix. The updated endpoint combines `sentiment_distribution` from articles with `reviews_sentiment_distribution` from reviews, producing a more realistic representative distribution of 89% positive, 7.65% negative, and 3.28% neutral. The dashboard therefore displays a user-friendly positive percentage rather than a raw compound VADER score, while still retaining per-keyword sentiment scores for trend interpretation.

D. Data Governance

The governance KPIs revealed systematic patterns consistent with the characteristics of all three data sources. For Spoonacular recipe data, the fields `preparationMinutes` and `cookingMinutes` exhibit null rates above 90%, which reflects that Spoonacular only populates these fields for recipes with separate preparation and cooking phases — a known API behavior rather than a data quality defect. For Eater NY articles and OpenTable reviews, the NLP compression ratio confirmed effective cleaning: summaries and review texts were reduced to semantically dense representations without loss of gastronomic signal, as validated by keyword extraction quality. The new sources endpoint also exposes article counts, article links, total review counts, and review volume by restaurant, making data coverage visible to dashboard users. Schema compliance remained at 100% across all runs, confirming the robustness of the Silver processing logic.

E. Consumption Layer

All three consumer interfaces — the Dash dashboards, the Next.js web application, and the Expo mobile application — successfully consume the FastAPI layer. The Governance dashboard provides the data engineering team with a real-time view of pipeline health, while the Storytelling dashboard delivers trend insights in a format accessible to non-technical users (chefs, event planners). The web trend page was redesigned to show the positive sentiment percentage, a readable mood label, keyword labels instead of raw TF-IDF scores, and a donut chart with category percentages. The data dashboard was also reframed for end users with labels such as *Records collected*, *Missing data*, and *Data quality*, plus source cards that list Eater NY articles and OpenTable review counts by restaurant. The web application adds persistent authentication and a recipe favorites system, and the mobile application replicates the full dashboard experience in a portable format with offline favorites via secure local storage.

F. Discussion

The trend-driven pipeline design — where NLP analysis of editorial content and diner reviews dynamically determines what recipes are retrieved — is a key architectural distinction

over static keyword-based approaches. This coupling ensures that recipe recommendations remain aligned with current public discourse and real consumer feedback rather than fixed ingredient lists that may become stale.

The OpenTable integration mitigates the original editorial-bias limitation, but it also introduces dependency on a browser-driven scraper and an internal GraphQL response structure that may change over time. Future work should incorporate additional user-generated content sources, such as Yelp reviews or Reddit culinary forums, to reduce dependence on a single review platform. Additionally, the current keyword extraction uses unsupervised methods; supervised classification or topic modeling could improve precision for ambiguous terms. The system also currently operates through manual Airflow triggers; scheduled or near-real-time ingestion would benefit applications with stricter latency requirements, such as live event menu adjustments.

IV. CONCLUSIONS

This paper presented Empire’s Taste, a data pipeline that addresses a concrete gap in the hospitality industry: menu planning for events in competitive gastronomic environments like New York City has historically depended on chef intuition rather than systematic analysis of public food discourse and diner feedback. The system closes this gap by answering three specific questions for event professionals — what dishes and ingredients are trending in NYC right now, how diners and food media feel about them, and which recipes best match those trends while accommodating dietary restrictions. The central design decision was the trend-driven feedback loop between Silver keyword extraction and Bronze recipe ingestion: rather than querying a static ingredient list, the pipeline identifies the top food topics from Eater NY articles and OpenTable reviews via TF-IDF and uses them to dynamically shape the recipe corpus. The Workshop 4 implementation strengthened this design by adding browser-based OpenTable review ingestion, review-aware sentiment aggregation, source-coverage endpoints, and user-facing dashboard labels that translate technical scores into actionable menu-planning signals.

Future work should prioritize three improvements. First, expanding ingestion beyond Eater NY and OpenTable to additional NYC culinary sources such as Grub Street, Yelp, or Reddit would make the trend signal more robust on low-coverage days and reduce platform dependence. Second, replacing VADER with a domain-adapted transformer model would improve sentiment precision for the nuanced language of both food journalism and restaurant reviews. Third, evolving the Gold layer from a latest-run snapshot to a rolling multi-week aggregation window would smooth the trend signal and unlock longitudinal analyses of how gastronomic preferences shift across seasons — the most natural next step as the pipeline accumulates historical partitions in production.

ACKNOWLEDGMENTS

The authors thank Universidad Distrital Francisco José de Caldas for academic support.

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